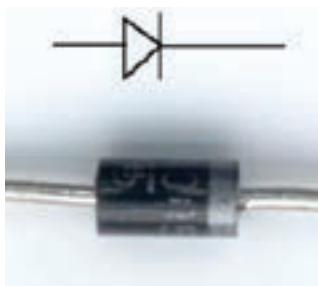


of quantities it can measure. However, for basic electronics and robotics purpose a simple and cheap multimeter would suffice. It will cost around ₹100. A snapshot of such a cheap and easy to use multimeter is given alongside. Also, the sections we would need to use frequently are highlighted.

2.2.5 Diodes

Diodes are two terminal devices just like the resistors, but they conduct electricity only in one direction. A resistor offers same resistance to the flow of electricity irrespective of its direction through it. On the other hand, a diode offers very large resistance to flow of electricity through it in one direction, and negligible resistance in other direction. Essentially, it means that a diode behaves as an open circuit (allowing no current flow) in one direction and as a closed circuit (allowing all the current to pass through it) in other direction. The condition of maximum current flow (or closed circuit) happens when the positive terminal of diode is connected to a point which is at higher potential than the point to which negative terminal is connected. This condition is called forward biasing. On the other hand, the condition of minimum or zero current flow happens when positive terminal of diode is connected to a point which is at lower potential than the point to which the negative terminal of diode is connected.

Before using any diode, you should always check if it is working fine by doing continuity test using a multimeter. The diode must offer continuous flow of current



A diode and its symbol

when its positive terminal is connected to red (positive) probe and its negative terminal is connected to black (negative) probe. The diode should offer discontinuity when multimeter is connected in opposite direction (that is Red probe to negative terminal of the diode and Black probe to positive terminal of the diode). Most commonly used diodes in electronics are 1N4001 and 1N4007.

2.2.6 LED

LEDs or light emitting diodes is a special category of diodes that emits light when forward biased. Generally, LEDs are of two types:

Visible light emitting LED (simple LED)